AGRONIX

Agricultural Governance & Agronomy Operating System

Official Field Operations, Statutory Enforcement & Agronomy Intelligence Platform — Mandal Agricultural Office (MAO), Tiryani, Kumram Bheem Asifabad District, Department of Agriculture, Government of Telangana.

Bilingual (English / Telugu) • Offline-First PWA • React 18 + TypeScript + Vite + Supabase

15+ Modules | Statutory PDF Engines | RAG Knowledge Base | Offline PWA + Android

1. About the Project

AGRONIX is an enterprise-grade, bilingual (English & Telugu), offline-first digital governance and agronomic management system engineered for grassroots agricultural administration in the Government of Telangana. Developed for the Mandal Agricultural Office (MAO), Tiryani — a predominantly tribal, remote agrarian jurisdiction — it bridges field inspection realities, statutory law enforcement, and real-time farmer advisory.

Agricultural extension officers manage regulatory, administrative and agronomic responsibilities across vast rural areas with unreliable connectivity. AGRONIX replaces disconnected paper registers and spreadsheets with a single digital workspace covering statutory enforcement, sample management, input dealer stock, subsidy schemes, and crop advisory.

2. Key Features

- Automated statutory legal engines — court-ready Show Cause Notices, Seizure Orders, Stop Sale Orders and Revocations with bilingual rendering, watermarking and strict clause citation.
- Confidential Sampling Hub — dual-coding registers, Fertilizer Form J/K/L, Pesticide Form XX/XXI and Seed Form VIII generation, covering letters with date lifecycle automation.
- Hybrid AI Knowledge Base (RAG) — pgvector + Ollama (bge-m3 embeddings, llama3.2) with cited answers from official Acts, Rules and Government Orders.
- On-device agronomic intelligence — crop disease detection, NPK fertilizer recommendations, tank-mix calculators and acreage converters that work fully offline.
- Supply chain & subsidy transparency — dealer stock reconciliation, IFMS validation and beneficiary tracking for NFSM and state seed allocations.
- Enforcement Tracker — end-to-end case management linking notices, samples, lab results and prosecution status.

3. Module Directory

Module	Purpose
Dashboard	Mandal overview, weather, mandi prices, maps and officer KPIs
Sampling Hub	Statutory forms, covering letters, dispatched-sample tracking
Officer Toolkit	Field tools, inspections, enforcement tracker, government portals
Saved Records & Storage	Officer records synced to cloud archive (R2 + Supabase)
Farmer Database	Farmer registry, Excel import, urea requirement analytics
Dealer Stock Portal	Fertilizer/seed/pesticide stock, receipts, sales and analytics
Farm Calculators	Acreage, fertilizer, pesticide, seed rate, irrigation, profitability, soil health
Crop Doctor / Intelligence	Crop advisory, protection recommendations, pest & disease guidance
Agrochemical Database	Pesticide catalogue, toxicity bands, CIBRC data
Knowledge Base	RAG document library, assistant, search and analytics (admin)
Documents	Forms library, GOs & circulars, file directory
Schemes	Subsidy tracking, farm mechanization, beneficiary records
Quality Control	QC targets and sample analysis tracking
Tour Diary	Monthly tour diaries, journeys, leaves and holidays
Contacts	Officer contact directory (AEO/MAO)

4. Technical Architecture

Frontend: React 18 + TypeScript SPA built with Vite, deployed on Vercel as an installable PWA with offline service worker and Android packaging via Capacitor. Backend: Supabase (PostgreSQL 17, Auth, Storage, Edge Functions in Deno) with Row-Level Security on all tables. AI services run server-side only — Ollama embeddings/LLM on a separate VPS, secrets held in Edge Function environment variables and never exposed to the browser. Generated legal documents are archived to Cloudflare R2 via a dedicated storage worker.

5. Security & Compliance

- Supabase Auth with email OTP verification, Cloudflare Turnstile bot protection, and strict password policy on registration/reset.
- RLS policies enforce admin/officer/dealer role separation; service-role keys exist only inside Edge Functions.
- Officer document archive uses client-side encryption and compression before cloud storage.

- Statutory documents carry watermark protection and bilingual (English/Telugu) rendering per departmental format.

6. Statutory & Legal Coverage

AGRONIX embeds the legal framework governing agricultural inputs in India:

- Essential Commodities Act, 1955 and FCO 1985 — fertilizer enforcement, stop-sale, seizure and revocation workflows.
- Seeds Act, 1966 — seed sample forms and licensing checks.
- Insecticides Act, 1968 — pesticide inspection, sample drawal and CIBRC formulation compliance.
- Fertiliser Control Order clauses, forms schedules and enforcement mind-maps built into the Officer Toolkit.

7. Deployment

The application is a static Vite SPA hosted on Vercel with SPA rewrites and cache headers. Database migrations are applied via the Supabase CLI (supabase db push) and Edge Functions deployed with supabase functions deploy. An Android build is produced through Capacitor for field officers requiring a native install.

8. Project Repository Layout

Path	Contents
project/src/app	Router, page switcher, layout shell
project/src/features	Feature modules (dashboard, sampling-hub, officers-toolkit, calculators, knowledge-base, etc.)
project/src/shared	UI kit, contexts, Supabase client, utilities
supabase/migrations	PostgreSQL schema migrations
supabase/functions	Deno Edge Functions (OTP, RAG, ogd-proxy, account deletion)
workers/	Cloudflare storage worker for officer records
docs/	Architecture and project documentation